

Posttraumatic Stress Symptoms in Australian Parents of Autistic Children: Factor Structure of the International Trauma Questionnaire (ITQ) and the Posttraumatic Stress Disorder Checklist for the *Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition* (PCL-5)

Kylie Hinde^{1, 2}, Gert Martin Hald^{1, 2}, David Hallford¹, John Gilmour³, Mikkel Arendt⁴,
Silvia Pavan¹, and David Austin^{1, 2}

¹ Faculty of Health, Deakin University

² Department of Public Health, University of Copenhagen

³ School of Social Science, University of Queensland

⁴ Department of Clinical Medicine, Aarhus University

Objective: Research indicates that parents of autistic children may have a higher risk of posttraumatic stress disorder (PTSD) than parents of neurotypical children. This study was to determine the optimal factor structure of two trauma screening assessments: the PTSD Checklist for *Diagnostic and Statistical Manual of Mental Disorders, fifth edition* (DSM-5) and the International Trauma Questionnaire, within this parent population. **Method:** Confirmatory factor analysis examined and compared one novel DSM-5 model and 14 previously identified DSM-5 and *International Classification of Diseases 11th Revision* (ICD-11) trauma symptom models among Australian parents of autistic children ($N = 563$). **Results:** Three DSM-5 PTSD models (anhedonia, hybrid, and intrusion/distress) provided a marginal fit, with the novel intrusion/distress model offering a superior fit. None of the five DSM-5 models tested achieved a good overall fit. For ICD-11, a three-factor model best fit the latent structure of PTSD symptoms. For complex PTSD (CPTSD), a two-factor second-order model and a six-factor first-order model provided a superior fit over five alternative CPTSD models. **Conclusions:** Results strongly support the internal reliability and construct validity of the ICD-11 PTSD and CPTSD models in Australian parents of autistic children, as measured by the International Trauma Questionnaire. The ICD-11 models outperformed the best DSM-5 models, highlighting their superiority for this population. While more complex DSM-5 models showed better fit than simpler ones, they still did not achieve a good overall fit. Notably, the results were largely consistent when assessing individuals meeting Criterion A for PTSD, specifically in relation to parenting-related traumatic experiences. The International Trauma Questionnaire is a more reliable and suitable tool for assessing PTSD and CPTSD in Australian parents of autistic children.

Clinical Impact Statement

This study highlights the superior reliability and validity of the *International Classification of Diseases 11th Revision* models in assessing posttraumatic stress disorder (PTSD) and complex PTSD in Australian parents of autistic children, compared with *Diagnostic and Statistical Manual of Mental Disorders, fifth edition* models. The findings support the use of the International Trauma Questionnaire as a more suitable tool for this population, capturing the distinct traumatic stress symptoms they experience. Results indicate that *International Classification of Diseases 11th Revision* models, particularly the three-factor PTSD and six-factor complex PTSD models, provide a better fit and clinical utility. These insights underscore the need for trauma-informed care tailored to the unique experiences of parents in this population.

Keywords: posttraumatic stress disorder, complex posttraumatic stress disorder, confirmatory factor analysis, autism

Supplemental materials: <https://doi.org/10.1037/tra0001959.supp>

This article was published Online First June 23, 2025.

Paul Frewen served as action editor.

Kylie Hinde  <https://orcid.org/0000-0001-8707-1618>

This article was funded by Deakin University, Københavns Universitet.

The authors have used identity-first language (e.g., “autistic child”) rather than person-first language (e.g., “child with autism”). This choice aligns with the neurodiversity-affirming movement, which seeks to de-stigmatize autism, as well as research demonstrating that most autistic individuals prefer

identity-first terminology (Botha et al., 2024; Bury et al., 2023; Izuno-Garcia et al., 2023; Taboas et al., 2023).

Open Access funding provided by Deakin University: This work is licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0; <https://creativecommons.org/licenses/by/4.0>). This license permits copying and redistributing the work in any medium or format, as well as adapting the material for any purpose, even commercially.

Kylie Hinde played a lead role in conceptualization, data curation,

continued

Posttraumatic stress disorder (PTSD) and its companion diagnosis, complex PTSD (CPTSD), are trauma- and stressor-related disorders that are precipitated by exposure to a traumatic event or series of events (World Health Organisation, 2018). PTSD is characterized by symptoms grouped into four clusters: intrusion, avoidance, negative alterations in cognition and mood, and alterations in arousal and reactivity. In contrast, CPTSD encompasses the core PTSD symptoms along with additional disturbances in self-organization (DSO), which include affect dysregulation, negative self-concept, and difficulties in interpersonal relationships (Cloitre et al., 2019). The two major international diagnostic classification systems for PTSD (the *Diagnostic and Statistical Manual of Mental Disorders, fifth edition—text rev.* [DSM-5-TR] and the *International Classification of Diseases 11th Revision* [ICD-11]) differ in their conceptualization of these disorders. The DSM-5-TR focuses solely on PTSD and does not include CPTSD as a distinct diagnosis. Instead, it incorporates symptoms related to DSO within its alternative model for personality disorders. In contrast, the ICD-11 explicitly distinguishes between PTSD and CPTSD, treating them as related but separate conditions. Evidence indicates that diagnostic rates of PTSD may vary depending on which system is used and with which population and type of trauma exposure (Elliott et al., 2021). A number of existing self-report measures can be used to screen for PTSD based on these systems. The PTSD Checklist for DSM-5 (PCL-5) and the International Trauma Questionnaire (ITQ) have been used extensively to assess PTSD according to DSM and ICD criteria respectively in a number of parent samples (e.g., Karatzias et al., 2023; Martin et al., 2021; Rahimi et al., 2023; Stewart et al., 2020; Sveen et al., 2016; Wang et al., 2019; Xiong, McGrath, Stewart, et al., 2022; Xiong, McGrath, Yakovenko, et al., 2022).

Despite widespread use, no studies to date have examined the PCL-5 or ITQ factor structures in parent samples and more specifically in parents of autistic children who are known to experience higher rates of traumatic stress symptomatology than parents of neurotypical children (Xiong, McGrath, Yakovenko, et al., 2022). The elevated rates of PTSD and CPTSD among this parent population underscore the profound impact of cumulative parenting-related traumatic stressors. Such experiences, which often include managing sensory sensitivities, communication differences, and aggressive behaviors (e.g., self-harm, suicidality, and absconding), are exacerbated by social isolation, financial strain, and inadequate support services (Stephenson et al., 2023; Xiong, McGrath, Yakovenko, et al., 2022). These challenges highlight the critical need for valid and reliable assessment tools to identify trauma in this population and guide appropriate interventions and support. These unrelenting and cumulative stressors heighten vulnerability to both PTSD and CPTSD, particularly in the context of ongoing rather than single traumatic events. Despite the absence of specific investigations into CPTSD prevalence within this population, the unrelenting and multifaceted nature of their stressors suggests that

this area warrants further exploration. By including CPTSD in this study, we hope to support this line of investigation.

Investigation into the factor structure of PTSD and CPTSD in parents of autistic children is needed to build evidence for the clinical utility of self-report measures for assessing traumatic stress symptoms within this unique population. As detailed below, existing research indicates that the factor structure of both the PCL-5 and ITQ vary across populations, and as such, their appropriateness for effectively measuring these symptoms across different populations cannot be assumed. If DSM-5 and ICD-11 factor models show a poor fit in samples of parents of autistic children, then the PCL-5 and ITQ measures may be mischaracterizing how PTSD and/or CPTSD symptoms present in these parent populations. This area of inquiry has implications for ongoing research in the area, assessment and treatment of PTSD and CPTSD in clinical settings, parent access to appropriate support and intervention, and child and family overall well-being.

DSM-5 PTSD Models

A PTSD diagnosis based on the DSM-5-TR requires exposure to a traumatic event (“Criterion A”), and endorsement of symptoms from four core symptom clusters: reexperiencing (Criterion B), avoidance (Criterion C), negative alterations in cognition and mood (Criterion D), and hyperarousal (Criterion E). Numerous studies have examined the latent structure of DSM-5 PTSD with at least several competing models proposed. The four-factor model (DSM-5 Model 1) is composed of the four symptom clusters mentioned above and provides the basis for all subsequent DSM-5 models (S. Murphy et al., 2018). Subsequent model iterations include the externalizing behavior model, which separates the core alterations in arousal and reactivity symptom cluster into three additional symptom clusters (externalizing behavior, anxious arousal, dysphoric arousal) to emphasize symptoms relating to distress or dysregulation (Armour et al., 2016); the anhedonia model, which identifies two additional distinct symptom clusters relating to negative affect and anhedonia (seven clusters in total; Lee et al., 2019). Anhedonia, or the diminished ability to experience pleasure, is particularly relevant in trauma-related disorders as it reflects the profound impact of trauma on positive emotional experiences and engagement with previously enjoyable activities (Lee et al., 2019). Last, the hybrid model, which emphasizes a more holistic PTSD model, is composed of a combination of seven symptom clusters drawn from the aforementioned models (Gilmour & Romaniuk, 2020). The fit of these models depends on the type of traumatic event(s) experienced, co-occurring conditions, and the sample characteristics in any given population (Armour et al., 2016; Seo & Cho, 2023). See Table 1 for the operationalization of the DSM-5 models depending on the type of trauma experienced, the characteristics of the samples, and other conditions.

investigation, methodology, and writing—original draft. Gert Martin Hald played a supporting role in supervision, writing—original draft, and writing—review and editing and an equal role in investigation and methodology. David Hallford played a supporting role in formal analysis, investigation, and supervision and an equal role in conceptualization, methodology, and writing—review and editing. John Gilmour played a supporting role in formal analysis and methodology. Mikkel Arendt played a supporting role in supervision and an equal

role in writing—review and editing. Silvia Pavan played a supporting role in project administration and writing—review and editing. David Austin played a lead role in data curation and supervision and an equal role in methodology, project administration, writing—original draft, and writing—review and editing.

Correspondence concerning this article should be addressed to Kylie Hinde, Faculty of Health, Deakin University, 221 Burwood Highway, Burwood, VIC 3125, Australia. Email: khinde@deakin.edu.au

Table 1
Item Mappings Across the DSM-5 and ICD-11 PTSD Models

DSM-5 symptom	DSM-5 model				ICD-11 model		
	DSM-5 (Model 1) ^a	Externalizing behavior (Model 2) ^a	Anhedonia (Model 3) ^a	Hybrid (Model 4) ^a	Intrusion/distress (Model 5) ^a	Three-factor (Model 6) ^b	One-factor (Model 8) ^b
B1. Intrusive thoughts	Intrusions	Intrusions	Intrusions	Intrusions	Intrusions	Reexperiencing	PTSD
B2. Nightmares	Intrusions	Intrusions	Intrusions	Intrusions	Intrusions	Reexperiencing	PTSD
B3. Flashbacks	Intrusions	Intrusions	Intrusions	Intrusions	Intrusions	Reexperiencing	PTSD
B4. Emotional cue reactivity	Intrusions	Intrusions	Intrusions	Intrusions	Distress reactivity		
B5. Physiological cue reactivity	Intrusions	Intrusions	Intrusions	Intrusions	Distress reactivity		
C1. Avoidance of thoughts	Avoidance	Avoidance	Avoidance	Avoidance	Avoidance	Avoidance	PTSD
C2. Avoidance of reminders	Avoidance	Avoidance	Avoidance	Avoidance	Avoidance	Avoidance	PTSD
D1. Trauma-related amnesia	NACM	NACM	Negative affect	Negative affect	Negative affect		
D2. Negative beliefs	NACM	NACM	Negative affect	Negative affect	Negative affect		
D3. Distorted blame	NACM	NACM	Negative affect	Negative affect	Negative affect		
D4. Pervasive negative emotional state	NACM	NACM	Negative affect	Negative affect	Negative affect		
D5. Lack of interest	NACM	NACM	Anhedonia	Anhedonia	Anhedonia		
D6. Feeling detached	NACM	NACM	Anhedonia	Anhedonia	Anhedonia		
D7. Inability to experience positive emotions	NACM	NACM	Anhedonia	Anhedonia	Anhedonia		
E1. Irritability/aggression	Arousal/reactivity	Externalizing beh	Dysphoric arousal	Externalizing beh	Externalizing beh		
E2. Recklessness	Arousal/reactivity	Externalizing beh	Dysphoric arousal	Externalizing beh	Externalizing beh		
E3. Hypervigilance	Arousal/reactivity	Anxious arousal	Anxious arousal	Anxious arousal	Anxious arousal	Perceived threat	PTSD
E4. Exaggerated startle	Arousal/reactivity	Anxious arousal	Anxious arousal	Anxious arousal	Anxious arousal	Perceived threat	PTSD
E5. Difficulty concentrating	Arousal/reactivity	Dysphoric arousal	Dysphoric arousal	Dysphoric arousal	Dysphoric arousal		
E6 Sleep disturbances	Arousal/reactivity	Dysphoric arousal	Dysphoric arousal	Dysphoric arousal	Dysphoric arousal		

Note. DSM-5 = *Diagnostic and Statistical Manual of Mental Disorders, fifth edition*; ICD-11 = *International Classification of Diseases 11th Revision*; PTSD = posttraumatic stress disorder; B = intrusion symptom; C = avoidance symptom; D = negative alterations in cognitions and mood symptom; NACM = negative alterations in cognitions and mood; E = alterations in arousal and reactivity; Externalizing beh = externalizing behavior.

^a Armour et al. (2016), Gilmour and Romaniuk (2020), Hansen et al. (2017), Krüger-Gottschalk et al. (2022), Lee et al. (2019). ^b Cloitre et al. (2014), Elklit et al. (2014), Glück et al. (2016), Islam et al. (2021), Hansen et al. (2015, 2017), Hyland et al. (2017), Karatzias et al. (2016), Maercker et al. (2013).

Since the reformulation of the *DSM* PTSD criteria as part of the revisions made for the 5th edition, confirmatory factor analysis (CFA) has generally demonstrated a relatively poor fit of the default four-factor model (*DSM-5* Model 1) using self-report assessments across various populations. The poor fit of this model is even more exaggerated when comparing subsequent models that consist of subdivisions of the four core PTSD symptom clusters (Blevins et al., 2015; Bovin et al., 2016; Wortmann et al., 2016). In particular, both the anhedonia and hybrid models, in comparison to other existing *DSM-5* models, demonstrate a better fitting PTSD model across various sample populations (Armour et al., 2015, 2016; Blevins et al., 2015; Forbes et al., 2015; Hansen et al., 2021; Islam et al., 2021; Krüger-Gottschalk et al., 2022; Soberón et al., 2016; Yang et al., 2017). While this may indicate that complex or multifactorial PTSD models may offer a superior fit over unidimensional *DSM-5* models across a variety of samples, it may also suggest that existing measurement tools do not capture how the *DSM* specifies PTSD (Soberón et al., 2016).

From a conceptual standpoint, there also seems to be some room for a finer distinction in measurement. For example, when considering the items that account for the PTSD criterion B (reexperiencing symptoms), it seems plausible that an additional model lies in the splitting of these items. More specifically, it may be argued that symptoms B1 to B3 (B1—distressing memories, B2—distressing dreams, and B3—flashbacks) may represent a stimulus and that items B4 and B5 (B4—psychological distress and B5—physiological reactions) better represent a distress reaction to the stimulus. This distinction has not been explored in existing factorial research.

ICD-11 PTSD Models

An *ICD-11* PTSD diagnosis requires exposure to a traumatic event (“Criterion A”), however, differs from the *DSM-5-TR* system in that endorsement of one or two symptoms from only three core symptom clusters is required (reexperiencing, avoidance, and sense of current threat), with the addition of functional impairment. Multiple studies have investigated the latent structure of PTSD according to the *ICD-11* criteria. The ITQ has been utilized to examine the latent structure of PTSD in persons exposed to natural disasters, accidents, sexual assault, physical or mental violence, life-threatening illness, and sudden accidental death (Hansen et al., 2015, 2021; Karatzias et al., 2016; Somma et al., 2019; Vallières et al., 2018). Specifically, three models of the PTSD criteria have been developed and tested, with the operationalization of these models described in Table 1. Although all three models have demonstrated a reasonably good fit across a number of samples, there has been some variation in the best-fitting model. For example, Armour et al. (2021), Hansen et al. (2017), and Krüger-Gottschalk et al. (2022), respectively, found that while all three *ICD-11* PTSD models demonstrated a good fit to the data as tested by the ITQ measurement scale, the three-factor model resulted in a superior fit in comparison to the two other models. Goodness-of-fit was attributed to a combination of sample composition, trauma type, and measurement method. In contrast, Glück et al. (2016) reported the one-factor model as having an excellent fit in a sample of elderly survivors of childhood trauma, and both Forbes et al. (2015) and Haravuori et al. (2016) reported the two-factor model as the best fitting of the three models due to theoretical coherence with the clinical characteristics of their respective samples (survivors or traumatic injury and people exposed to mass shootings). Other studies that tested the three-factor

model without comparison to the other models demonstrated its excellent fit to the data across a variety of populations (Hansen et al., 2015; La Greca et al., 2017; Tay et al., 2015).

For instance, Hansen et al. (2015) applied the model using the Harvard Trauma Questionnaire in a Danish population, though this approach adheres to *DSM-IV* criteria only, rather than assessing current *DSM-5* and *ICD-11* frameworks. La Greca et al. (2017) tested the model in children and adolescents affected by hurricanes. This focuses on event-specific trauma within a specific developmental and cultural context and therefore may not be generalizable. Tay et al. (2015) evaluated the model in refugee populations, which may not be generalizable nor reflect the chronic parenting-related stressors explored here. Given these methodological and contextual differences in the present study, they are not able to meaningfully inform our research question. Rather, our focus on testing PTSD/CPTSD models in the context of a sample of parents of autistic children offers a unique and valuable contribution to the field.

Comparison of DSM-5 and ICD-11 PTSD Models

Few studies to date have examined the latent structure of the two PTSD diagnostic systems in the same samples. Those that have been conducted indicate that there may be implications for choosing one diagnostic system over the other (Elliott et al., 2021; Hansen et al., 2017; Krüger-Gottschalk et al., 2022; Tay et al., 2015). For example, Hansen et al. (2017) found significantly higher rates of PTSD with the PCL-5 in comparison to the ITQ for a university student sample and almost double the rate of PCL-5 PTSD to that of the ITQ in a veteran sample. Furthermore, the results indicated that the three-factor *ICD-11* model provided a satisfactory fit across all sample populations, whereas the *DSM-5* model provided a marginally acceptable fit and was the poorest fitting of all the models tested (Hansen et al., 2017). Other studies have evaluated the fit of the *DSM-5* and *ICD-11* using different measurement tools (e.g., Krüger-Gottschalk et al., 2022; Tay et al., 2015), and generally indicate greater support for the *ICD-11* factor structure compared with the *DSM-5*, regardless of the measurement used. These studies attributed the *ICD-11*’s superior fit to its greater parsimony in form, reduced item variation, and focus on fewer constructs. Collectively, existing findings indicate possible risks of overestimating or underestimating PTSD and CPTSD symptoms (and/or diagnosis) depending on the diagnostic criteria and measurement tool utilized in both clinical and research spaces. For example, overestimation may occur when tools like the PCL-5 capture a broader range of symptoms than those defined under *ICD-11*, potentially inflating PTSD prevalence rates in certain populations. Conversely, underestimation may arise if *ICD-11*’s narrower focus on fewer constructs overlooks meaningful symptomatology that impacts functioning but falls outside its diagnostic criteria. These discrepancies have significant implications for accurately identifying and treating trauma-related disorders, particularly in parenting populations, where unique stressors may challenge traditional definitions of trauma and symptom thresholds.

ICD-11 CPTSD Models

For a diagnosis of CPTSD, the *ICD-11* requires endorsement of the three main PTSD symptom clusters (reexperiencing, avoidance, and sense of current threat), as well as the additional endorsement of

DSO symptom clusters (affective dysregulation, negative self-concept, disturbances in relationships) and functional impairment. Several studies have examined the latent structure of CPTSD according to *ICD-11* criteria. Seven different models have been proposed with the majority of existing studies using the ITQ to examine the CPTSD latent structure. These models are described in Table 2.

A systematic review of 33 factor analytic studies indicated that of these seven CPTSD models, both a six-factor model and a two-factor second-order model best represented the latent structure of the ITQ (Redican et al., 2021). In addition, evidence indicates that both models provide an optimal fit across various sample populations, with this possibly attributed to how the models capture the distinction between PTSD and DSO symptoms inherent in the formulation of PTSD and CPTSD (World Health Organisation, 2018). For example, both models demonstrated a good fit for a sample of survivors of sexual violence in Ireland (Frost et al., 2022), in prison staff (Dhingra et al., 2021), in veterans and combat soldiers (Armour et al., 2021; Mordeno et al., 2019; D. Murphy et al., 2020), and in refugees (Silove et al., 2017).

In light of existing literature that describes the prolonged, repetitive nature of parenting-related traumatic events experienced by some parents of autistic children (Stewart et al., 2020), CPTSD clinical conceptualization (as measured by the ITQ) is likely to be present in some or many cases. Therefore, further examination of the construct validity in self-report of related symptoms is warranted in this parent population.

The Present Study

The aim of the present study was to evaluate the fit of a number of *DSM-5* and *ICD-11* factorial models of both PTSD (*DSM* and *ICD*) and CPTSD (*ICD*) in a sample of parents of autistic children, as measured by the PCL-5 and the ITQ. These measures were selected due to their robust psychometric properties, their alignment with *DSM-5* and *ICD-11* diagnostic frameworks, and their suitability for online administration. The PCL-5 is well-established for assessing *DSM-5* PTSD symptoms, while the ITQ is specifically designed for *ICD-11* PTSD and CPTSD criteria. To assess Criterion A (exposure to traumatic events), the Life Events Checklist for *DSM-5* (LEC-5) was used. The LEC-5 was chosen for its brevity, validation in trauma research, and compatibility with the PCL-5. Alternative tools, such as the Trauma History Questionnaire or the Posttraumatic Diagnostic Scale–5, were excluded due to concerns about length and suitability for an online survey format. In addition to analysis of the full sample, a subset analysis was conducted on participants who met Criterion A, a prerequisite for PTSD diagnosis, influenced the fit of the tested models. The results of this analysis provide additional insights into the applicability of *DSM-5* and *ICD-11* models for PTSD and CPTSD in this population.

A novel aspect of this study was that the responses to the PCL-5 and ITQ were in relation to self-report parenting-specific experiences, and not in response to generic traumatic stressors listed in standardized measurement scales such as the LEC-5. No studies have examined the *DSM-5* and/or *ICD-11* model fit in parents of autistic children, therefore, the *DSM-5* and *ICD-11* model selection for this study was based on the findings of previous studies on a variety of different community and clinical populations. However, it

Table 2
Item Mappings Across the *ICD-11* CPTSD Models

ICD-11 symptom	ICD-11 model ^a					
	Unidimensional (Model 1)	Two-factor first-order (Model 2)	Six-factor first-order (Model 3)	Single-factor second-order (Model 4)	Two-factor second-order (Model 5)	Two-factor second-order with PTSD (Model 6)
P1. Nightmares	CPTSD	PTSD	Reexperiencing	Re-CPTSD	Re-PTSD	Re-PTSD
P2. Flashbacks	CPTSD	PTSD	Reexperiencing	Re-CPTSD	Re-PTSD	Re-PTSD
P3. Avoidance of thoughts	CPTSD	PTSD	Avoidance	Av-CPTSD	Av-PTSD	Av-PTSD
P4. Avoidance of reminders	CPTSD	PTSD	Avoidance	Av-CPTSD	Av-PTSD	Av-PTSD
P5. Hypervigilance	CPTSD	PTSD	Threat	Th-CPTSD	Th-PTSD	Th-PTSD
P6. Exaggerated startle	CPTSD	PTSD	Threat	Th-CPTSD	Th-PTSD	Th-PTSD
C1. Difficulty concentrating	CPTSD	DSO	Affective dysregulation	AD-CPTSD	AD-PTSD	AD-PTSD
C2. Sleep disturbances	CPTSD	DSO	Affective dysregulation	AD-CPTSD	AD-PTSD	AD-PTSD
C3. Feel like a failure	CPTSD	DSO	Negative self-concept	NSC-CPTSD	NSC-DSO	NSC-DSO
C4. Feeling worthless	CPTSD	DSO	Negative self-concept	NSC-CPTSD	NSC-DSO	NSC-DSO
C5. Feeling distant or cutoff	CPTSD	DSO	Disturbance in relationships	DR-CPTSD	DR-DSO	DR-DSO
C6. Hard staying emotionally close	CPTSD	DSO	Disturbance in relationships	DR-CPTSD	DR-DSO	DR-DSO

Note. *ICD-11* = International Classification of Diseases 11th Revision; CPTSD = complex posttraumatic stress disorder; P = PTSD; PTSD = posttraumatic stress disorder; Re = reexperiencing thoughts; Av = avoidance; Th = sense of threat; C = CPTSD; DSO = disturbances in self-organization; AD = affective dysregulation; NSC = negative self-concept; DR = disturbances in relationships.
^a Armour et al. (2021), Gilbar et al. (2018), Hansen et al. (2017), Hyland et al. (2017), Karatzias et al. (2016), Mordeno et al. (2019), S. Murphy et al. (2017), Shevlin et al. (2017), Silove et al. (2017), Tay et al. (2018).

is hypothesized that the factor structures may differ in this specific population due to the unique and chronic nature of stress associated with parenting an autistic child. Unlike trauma related to isolated events, these parents face ongoing caregiving demands, social isolation, and the emotional toll of managing their child's complex needs. These factors could contribute to a distinct composition of PTSD and CPTSD symptoms, potentially leading to differences in the way symptoms cluster compared with other traumatized populations. Additionally, the frequent overlap with other mental health issues, such as anxiety and depression, may further influence symptom patterns, warranting exploration of alternative factor structures.

In addition, one novel *DSM-5* model (intrusion/distress) was tested and this was based on the hypothesis that criterion B (reexperiencing symptoms) may be more accurately represented by two distinct symptom types: (a) psychopathological alterations in mental and behavioral functioning (intrusive memories and distressing dreams) and (b) psychological and physiological dysregulation (dissociative reactions and psychological distress). Four other well-established *DSM-5* models were tested (*DSM-5*, externalizing behavior, anhedonia, and the hybrid model). Three *ICD-11* PTSD models were tested (one-, two-, and three-factor models), and seven CPTSD models were tested.

Method

Procedure and Participants

An online cross-sectional survey was conducted in Australia targeting parents of autistic and neurotypical children. Eligibility required parents to be at least 18 years old, residing in Australia, with a child aged 17 or younger. This age limit was chosen to focus on parents actively engaged in caregiving roles, as the demands of parenting are particularly acute during childhood and adolescence. This stage often involves heightened stressors, especially for parents of autistic children, related to education, behavior, and health care needs (Stephenson et al., 2023).

Parents of autistic children were required to confirm in the online survey whether their child had received a formal autism diagnosis by answering "yes" or "no." Participants were also asked to specify the type of autism diagnosis (e.g., Autism Spectrum Disorder, Asperger's Disorder) and the child's age at the time the diagnosis was made. Data from parents of neurotypical children were collected as part of a larger study, where they served as a control group. They are not reported on in this article as the research aims and analyses do not require a control group. Participants were recruited nationwide through various methods, including emails to autism organizations, social media ads, online forums, support groups, health clinics, and community notice boards. Snowball sampling was also employed, allowing participants to share the survey with other parents. Participants did not receive monetary compensation for their participation; however, they were given the option to enter a prize draw to win one of four noise-canceling headsets.

Participants were eligible for inclusion in the study if they resided in Australia, were the parent of a child aged 17 years or younger, and had a child with a formal autism diagnosis if participating as a parent of an autistic child. Participants were excluded if they did not reside in Australia, had a child older than 17 years, did not have a child with a formal autism diagnosis (if participating as a parent of an autistic child), had a history of severe cognitive impairment or neurological

disorder that could impact questionnaire comprehension, or had a history of significant substance use disorder that could interfere with symptom reporting. Participants completed three questionnaires (LEC-5, ITQ, and PCL-5) in a fixed order: LEC-5 first, followed by ITQ, and then PCL-5. This order ensured that participants anchored their responses to a specific traumatic event identified in the LEC-5. Participants were instructed to refer to the same event for both the ITQ and PCL-5. While the possibility of order effects cannot be ruled out, this sequence minimized ambiguity and maintained clarity in symptom reporting.

This study composed of 563 parents of autistic children. On average, parents were around 43.1 years old, with children averaging 11.5 years. The majority of parents were female (84.6%), and mothers (84.4%), and a significant proportion were in married or domestic partnerships (72.6%). Household composition varied, with 67.1% having both biological parents, while 28.4% were single biological parents. The ethnic makeup was primarily Caucasian or white European (88.9%). A notable portion of the parents held university degrees or higher (54.4%), and employment was distributed across full-time (29.7%), part-time (36.1%), and stay-at-home parenting (22.3%). In terms of income, 42.1% reported earnings over \$100,000, while 31.3% had less than \$74,000. Most families had two children (49.2%) in the household. Comprehensive sample characteristics are described in Table 3.

Participants completed the measures for this study as part of a broader questionnaire investigating parenting-related PTSD and CPTSD. Participants completed part two of the LEC-5, in which they reported their most challenging parenting-related event/s (traumatic event) and additional questions relating to this event. The participants then completed the ITQ and PCL-5 along with other measures. Across the entire sample, a total of 36.2% ($n = 204$) of participants reported events that met Criterion A for PTSD, as defined by exposure to actual or threatened death or serious injury (*DSM-5*), or extremely threatening or horrific events (*ICD-11*). Conversely, 63.8% ($n = 359$) of participants did not meet Criterion A.

Measures

LEC-5 Part Two

The LEC-5 (Part 2) was used to assess participants' exposure to potentially traumatic parenting-related experiences (Weathers et al., 2013). The wording was modified from the original version of the measure such that the instruction, "briefly describe the worst event," was instead presented as "briefly identify the worst event in parenting your autistic child." This was done to orient participants to their most challenging parenting experience rather than any other nonparenting experiences. The LEC-5 (Part 2) used in this study consisted of seven items that asked the participant to first give a brief description of a very stressful parenting event/s, when the event occurred, how it was experienced, if the event involved actual or threatened death, serious injury, and number of exposures to similar events. The item on sexual violence was excluded due to the absence of evidence of such incidents by autistic children toward parents, streamlining the survey to reduce participant burden. This exclusion is discussed as a possible limitation. No other adaptations to the instructions or to the LEC-5 items were made.

As the LEC-5 part two is a descriptive checklist, not a psychometric scale, traditional reliability measures like Cronbach's α

Table 3
Sociodemographic Characteristics of the Parent Sample

Characteristic	<i>M</i>	<i>SD</i>	Range	<i>N</i>	%
Parent age (years)	43.1	6.6	21–61	563	
Child age (years)	11.5	3.9	2–17	563	
Parent gender					
Female				473	84.6
Male				80	14.3
Other				6	1.1
Child gender					
Female				207	36.8
Male				326	57.9
Other				30	5.4
Parent relationship to child					
Mother				472	84.4
Father				80	14.3
Foster parent/other				7	1.3
Parent relationship status					
Married or domestic partnership				405	72.6
Separated or divorced				94	16.8
Other				59	10.6
Household composition					
Single biological parent				159	28.4
Both biological parents				375	67.1
Other				25	4.5
Ethnicity					
Caucasian or White European				497	88.9
Asian				20	3.6
Other				42	7.5
Highest education					
University degree or higher				278	54.4
Occupational/technical/vocational training				144	28.2
High school				67	13.1
Other				22	4.3
Current employment					
Full time				164	29.7
Part time				199	36.1
Stay-at-home parent				123	22.3
Unemployed and other				66	12
Household income (AUD)					
Over \$100,000				214	42.1
\$75,000 to \$99,999				102	20.1
Less than \$74,000				159	31.3
Prefer not to say				33	6.5
Number of children in family					
One				88	15.6
Two				277	49.2
Three				151	26.8
Four+				47	8.3

Note. *N* = 563.

are not applicable. However, the LEC-5 part two is widely validated for assessing trauma exposure in research and clinical settings (Weathers et al., 2013).

PTSD Checklist for DSM-5 (PCL-5)

The PCL-5 (Weathers et al., 2013) is a 20-item self-report measure assessing PTSD severity referencing the past month and was used to measure symptoms according to the *DSM-5* criteria of PTSD. The PCL-5 references four domains (reexperiencing, avoidance, negative alterations in cognition and mood, and hyperarousal) that are consistent with the four PTSD criteria of the *DSM-5*. For the purposes of this project, the original PCL-5 instructions were adapted to direct

parents to respond to “thinking about your worst event in parenting your autistic child you listed previously (i.e., from the LEC-5), which for this questionnaire means the event or events that currently bother you the most.” No other adaptations to the instructions or to the PCL-5 items were made. Participants rated the intensity of their symptom experience on a 5-point Likert scale (0 = *not at all*, 4 = *extremely*). In this study, PCL-5 demonstrated acceptable reliability ($\alpha = .79$).

ITQ

The ITQ (Cloitre et al., 2019) was used to measure symptoms according to the *ICD-11* criteria for PTSD and CPTSD. The ITQ PTSD and CPTSD subscales each comprise three core symptom clusters (PTSD: reexperiencing, avoidance, sense of threat; CPTSD: affective dysregulation, negative self-concept, disturbances in relationships) each assessed by two items. For the purposes of this project, the original ITQ instructions were adapted to direct participants to identify and report “a very stressful parenting experience/s with your autistic child ... and to indicate how much you have been bothered by that problem in the past month.” No other adaptations to the instructions or to the ITQ items were made. Participants rated the intensity of their symptom experience on a 5-point Likert scale (0 = *not at all*, 4 = *extremely*). In this study, the ITQ showed excellent reliability ($\alpha = .94$).

Data Analysis

Data screening was conducted in IBM SPSS Statistics (Version 29; IBM Corporation, Armonk, New York), and the CFA models were constructed and tested in IBM SPSS AMOS (Version 28). Little’s test indicated that less than 1% of the data was missing across both PCL-5 and ITQ scales, so mean replacement was used to replace all missing values. This approach was deemed appropriate due to the low proportion of missing data, which reduces the potential bias that could be introduced by more complex imputation methods such as multiple imputation. All item scores were within appropriate ranges. The item mappings of the four CFA models are shown in Tables 1 and 2. As the PCL-5 and *ICD-11* scores were found to have violations to the assumptions of univariate and multivariate normality, the Bollen–Stine bootstrap χ^2 test was used to assess significance of model fit (Bollen & Stine, 1992). In addition, the following global fit indices were used to determine model fit for each CFA: chi-square value (χ^2 ; Hayduk, 2014a), comparative fit index (CFI), root-mean-square error of approximation (RMSEA), and the standardized root-mean-square residual (SRMR). A good fit to the data was determined by a nonsignificant chi-square minimum (CMIN) value, a CMIN of ≤ 5 , a CFI of 0.95, and RMSEA and SRMR values of 0.08 or less (Hu & Bentler, 1999; Kline, 2022; Norris, 1990; West et al., 2012). Due to the exploratory nature of this research a slightly lower Cronbach’s α (.60) was deemed as an acceptable measure of internal reliability (Kline, 1999).

A number of the CPTSD models tested included models with a higher order factor that other factors loaded onto (e.g., six-factor first-order and the two-factor second-order models). The models were compared with determine the best-fitting models within each *ICD-11* and *DSM-5* classification system. The chi-squared difference test ($\Delta\chi^2$) was used to compare nested models and significant difference between models indicated that the more complex model represented a

better fit to the data. Bayesian information criteria (ΔBIC) differences were used to compare nonnested models in that lower Bayesian information criteria indicates a better fit and values of ≥ 10 indicates a poorer fit (Feigin et al., 2021).

In addition to analyzing the full sample ($N = 563$), a subset analysis was conducted on the 204 participants who met Criterion A for PTSD as defined by the *DSM-5* and *ICD-11* frameworks. This subset analysis was undertaken to evaluate model fit specifically within the group that met full criteria for PTSD and provide further insights into the factorial structure of PTSD and CPTSD symptoms.

Results

DSM-5 PTSD Models

All five *DSM-5* models were found to have statistically significant Bollen–Stine bootstrap χ^2 values ($p \leq .001$). The hybrid and intrusion/distress models were an adequate fit to the data, with CMIN < 5 , the CFI $> .95$, and the RMSEA $< .08$. The anhedonia model was marginally lower than the cutoff on the CFI. When comparing the externalizing behavior ($\Delta\chi^2 = 188.9, p \leq .001$), anhedonia ($\Delta\chi^2 = 346.94, p \leq .001$), hybrid ($\Delta\chi^2 = 434.79, p \leq .001$), and intrusion/distress ($\Delta\chi^2 = 471.82, p = .001$) models showed a significantly better fit to the data than the *DSM-5* model (Model 1). Of the two adequate fitting models, the intrusion/distress model showed a significantly better fit than the hybrid model ($\Delta\chi^2 = 37.03, p \leq .001$), suggesting that, of the *DSM-5* models, it provided a better fit to the data. As such, the goodness-of-fit for the intrusion/distress model is presented in Table 4.

For the intrusion/distress model, the items loaded significantly onto the respective first-order factors ranging from $r = .47$ to $r = .93$. In addition, the composite reliability estimates indicated acceptable reliability for all eight subscales in the intrusion/distress model:

intrusions = 0.88, distress/reactivity = 0.87, avoidance = 0.84, negative affect = 0.85, anhedonia = 0.90, anxious arousal = 0.83, and dysphoric arousal = 0.72, with the exception of externalizing behavior: 0.57. Factor loadings for the intrusion/distress model are presented in Table 5.

ICD-11 PTSD Models

The one-factor (Model 8) and second-factor (Model 7) *ICD-11* PTSD models demonstrated significant Bollen–Stine bootstrap χ^2 values ($p \leq .001$), CMIN values of > 5 , and RMSEA values of > 0.1 indicating an inadequate fit to the data. The three-factor model showed a good fit to the data across CFI, RMSEA, SRMR, and the Bollen–Stine bootstrap χ^2 tests. When comparing the *ICD-11* PTSD models, both the two-factor ($\Delta\chi^2 = 98.21, p \leq .001$) and three-factor models ($\Delta\chi^2 = 139.76, p \leq .001$) were a significantly better fit than the one-factor model. The three-factor model fit had an excellent fit to the data and showed a significantly better fit than the two-factor model ($\Delta\chi^2 = 41.55, p \leq .001$), suggesting that the three-factor model provides the best fit to the data. As such, the goodness-of-fit for the three-factor model is presented in Table 4.

For the three-factor model, the items loaded significantly onto the respective first-order factors ranging from $r = .71$ to $r = .86$. In addition, the composite reliability estimates indicated acceptable levels of internal reliability for the three subscales in the three-factor model: reexperiencing = 0.74, avoidance = 0.79, and sense of threat = 0.82. Factor loadings for the three-factor model are presented in Table 6.

ICD-11 CPTSD Models

The six-factor first-order model (Model 3) and the two-factor second-order model (Model 5) showed an excellent fit to the data

Table 4
Goodness-of-Fit Indices for the *DSM-5* and *ICD-11* PTSD and CPTSD Models

Model	Goodness-of-fit indices						
	χ^2	df	CMIN/df	CFI	RMSEA [90% CI]	SRMR	BIC
<i>DSM-5</i> PTSD							
1. <i>DSM-5</i>	952.43**	164	5.81	.902	.092 [.087, .098]	0.05	1243.76
2. Externalizing behavior	763.53**	155	4.93	.925	.084 [.078, .090]	0.05	1111.86
3. Anhedonia	605.49**	155	3.91	.944	.072 [.066, .078]	0.04	953.82
4. Hybrid	517.64**	149	3.47	.954	.066 [.060, .073]	0.04	903.97
5. Intrusion/distress	480.61**	142	3.39	.958	.065 [.059, .072]	0.04	911.28
<i>ICD-11</i> PTSD							
1. One-factor	151.75**	9	16.86	.909	.168 [.145, .192]	0.06	176.06
2. Two-factor	53.54**	8	6.69	.971	.101 [.076, .127]	0.03	135.88
3. Three-factor	11.99	6	1.99	.996	.042 [.000, .077]	0.01	106.99
<i>ICD-11</i> CPTSD							
1. Unidimensional	1141.08**	54	21.13	.692	.19 [.180, .199]	0.11	1293.08
2. Two-factor first-order	433.01**	53	8.17	.892	.11 [.103, .123]	0.05	591.34
3. Six-factor first-order	57.39	39	1.47	.995	.03 [.010, .044]	0.02	304.39
4. Single-factor second-order	365.19**	48	7.61	.910	.11 [.098, .119]	0.08	555.99
5. Two-factor second-order	78.95	47	1.68	.991	.04 [.021, .048]	0.02	275.28
6. Two-factor second-order with DSO	217.11**	50	4.34	.953	.08 [.067, .088]	0.03	394.44
7. Two-factor second-order with PTSD	294.30**	50	5.89	.931	.09 [.083, .104]	0.04	471.63

Note. Bolded models = best-fitting models; *DSM-5* = *Diagnostic and Statistical Manual of Mental Disorders, fifth edition*; *ICD-11* = *International Classification of Diseases 11th Revision*; PTSD = posttraumatic stress disorder; CPTSD = complex posttraumatic stress disorder; χ^2 = chi-square value; df = degrees of freedom; CMIN = chi-square minimum; CFI = comparative fit index; RMSEA = root-mean-square error of approximation; CI = confidence interval; SRMR = standardized root-mean-square residual; BIC = Bayesian information criteria; DSO = disturbances in self-organization.

** $p < .01$ (models significant).

Table 5*Standardized Factor Loadings and Factor Correlations for the Intrusion/Distress Model*

DSM-5 symptom	In	DR	Av	NA	An	EB	AA	DA
B1. Intrusive thoughts	.87							
B2. Nightmares	.81							
B3. Flashbacks	.84							
B4. Emotional cue reactivity		.88						
B5. Physiological cue reactivity		.87						
C1. Avoidance of thoughts			.86					
C2. Avoidance of reminders			.84					
D1. Trauma-related amnesia				.58				
D2. Negative beliefs				.79				
D3. Distorted blame				.79				
D4. Pervasive negative emotional state				.86				
D5. Lack of interest					.88			
D6. Feeling detached					.86			
D7. Inability to experience positive emotions					.85			
E1. Irritability/aggression						.73		
E2. Recklessness						.52		
E3. Hypervigilance							.81	
E4. Exaggerated startle							.87	
E5. Difficulty concentrating								.83
E6. Sleep disturbances								.67
Factor correlations								
Intrusions	—							
Distress/reactivity	.93	—						
Avoidance	.93	.93	—					
Negative affect	.79	.83	.83	—				
Anhedonia	.68	.70	.70	.82	—			
Externalizing behavior	.68	.70	.66	.91	.88	—		
Anxious arousal	.63	.67	.66	.70	.68	.72	—	
Dysphoric arousal	.55	.66	.47	.56	.63	.74	.64	—

Note. DSM-5 = *Diagnostic and Statistical Manual of Mental Disorders, fifth edition*; In = intrusions; DR = distress/reactivity; Av = avoidance; NA = negative affect; An = anhedonia; EB = externalizing behavior; AA = anxious arousal; DA = dysphoric arousal; B = intrusion symptom; C = avoidance symptom; D = negative alterations in cognitions and mood symptom; E = alterations in arousal and reactivity.

across CFI, RMSEA, SRMR, and the Bollen–Stine bootstrap χ^2 tests and as such are both presented in full in Table 4. When comparing the models with best fit indices, Model 5 provided a significantly better fit over Model 3 ($\Delta\chi^2 = 21.56, p = .002$). All other models demonstrated inadequate fit, with significant Bollen–Stine bootstrap χ^2 values ($p \leq .001$), CMIN values of ≥ 5 , and/or CFI of 0.95 or less (see Table 4). The chi-squared difference test indicated that all six multidimensional models were superior to the unidimensional model.

Regarding factor loadings, for Models 3 and 5, all items loaded significantly onto the respective first-order factors, however, in Model 5, the affective dysregulation factor showed a standardized indicator loading greater than 1.00. In Model 3, significant correlations were found between all first-order factors ranging from $r = 0.42$ to $r = 0.90$. In Model 5, all first-order factors loaded significantly onto the second-order factors and the correlation between PTSD and DSO factors was strong ($r = 0.99$). Factor loadings for Models 3 and 5 are presented in Table 6.

Additional cross-validation analyses were conducted by splitting the data set 70/30 (70% for training and 30% for testing) to assess the high correlations in Models 3 and 5, identify any overconstrained parameters, and detect any underlying statistical artifacts. In Model 3, the correlation between affective dysregulation and disturbances in relationships in the training set was 0.94 and 0.82 in the test set. This indicated a consistent performance across training, test, and original data sets, with all correlations falling below 1.00. In Model 5, the

correlation between affective dysregulation and DSO in the training set was 1.06 and 0.94 in the test set. While this indicates improved correlation in the test set, the training set, and original data set correlations still indicate multicollinearity or overfitting.

The composite reliability estimates across Models 3 and 5 indicated very similar and acceptable internal reliability for all subscales apart from the affective dysregulation subscale: reexperiencing = 0.74, avoidance = 0.82, sense of threat = 0.79 (Model 3) and 0.80 (Model 5), affective dysregulation = 0.58, negative self-concept = 0.90, and disturbed relationships = 0.83. For Model 5, the PTSD (0.92) and DSO (0.94) subscales had good internal reliability.

Goodness-of-Fit Analysis for Criterion A Subset ($n = 204$)

The results in this subset largely mirrored those obtained in the full sample, with some differences in goodness-of-fit indices. For the DSM-5 PTSD models, fit indices were generally better in the subset, with improved CMIN/ df and RMSEA values across all models. The intrusion/distress model demonstrated the best fit within the DSM-5 models in both samples, with CMIN/ $df = 2.14$, RMSEA = .075, 90% CI [.063, .086] for the subset, compared with CMIN/ $df = 3.39$, RMSEA = .065 in the full sample.

Similarly, the ICD-11 PTSD models showed improved fit in the subset, particularly for the three-factor model, which continued to

Table 6

Standardized Factor Loadings and Factor Correlations for the Three-Factor ICD-11 PTSD, Six-Factor, and the Two-Factor Second-Order CPTSD Models

ICD-11 symptom	Three-factor PTSD model			Six-factor first-order CPTSD model						Two-factor second-order CPTSD model					
	Re	A	PT	Re	A	PT	AD	NSC	DR	Re	A	PT	AD	NSC	DR
P1. Nightmares	.71			.72						.71					
P2. Intrusive memories/flashbacks	.82			.81						.82					
P3. Avoiding internal reminders		.80			.86						.86				
P4. Avoiding external reminders		.82			.81						.80				
P5. "Super-alert," watchful, on guard			.86			.79						.80			
P6. Feeling jumpy or easily startled			.81			.83						.83			
C1. Long time to calm down							.58						.57		
C2. Numb or emotionally shut down							.70						.71		
C3. Feel like a failure								.87						.87	
C4. Feeling worthless								.93						.93	
C5. Feeling distant or cutoff									.89						.89
C6. Hard staying emotionally close									.80						.80
Factor correlation										PTSD	DSO				
Reexperiencing thoughts	—			—						.96					
Avoidance	.86	—		.86	—					.88					
Threat	.79	.71	—	.79	.70	—				.82					
Affective dysregulation				.67	.59	.65	—				1.02				
Negative self-concept				.48	.42	.44	.83	—			.82				
Disturbances in relationships				.48	.49	.52	.90	.75	—		.89				

Note. All factor loadings and correlations are statistically significant ($p < .01$). ICD-11 = *International Classification of Diseases 11th Revision*; PTSD = posttraumatic stress disorder; CPTSD = complex posttraumatic stress disorder; Re = reexperiencing thoughts; A = avoidance; PT = threat; AD = affective dysregulation; NSC = negative self-concept; DR = disturbances in relationships; P = PTSD symptom; C = CPTSD symptom; DSO = disturbances in self-organization.

demonstrate the best fit overall (CMIN/ df = 1.78, RMSEA = .062, 90% CI [.000, .121]). This was consistent with its performance in the full sample, where it also emerged as the best-fitting model, albeit with slightly lower CMIN/ df and RMSEA values.

For the ICD-11 CPTSD models, results were more varied. While the six-factor first-order model remained the best-fitting model across both samples, its fit was slightly weaker in the subset (CMIN/ df = 2.00, RMSEA = .07) compared with the full sample (CMIN/ df = 1.47, RMSEA = .03). Conversely, the unidimensional model continued to show poor fit in both the subset and full sample. These findings suggest that model performance is generally consistent across the full sample and Criterion A subgroup, with slight improvements in fit indices in the subgroup for simpler models (e.g., DSM-5 PTSD and ICD-11 PTSD), but better performance in the full sample for more complex models (e.g., ICD-11 CPTSD). Further details on this analysis can be found in [Supplemental Table S1](#).

Discussion

This study was the first to assess the reliability and fit of several factorial PTSD and CPTSD models using the PCL-5 and the ITQ in a sample of parents of autistic children; a population known to experience higher rates of traumatic parenting experiences and symptoms than parents of typically developing children (Stewart et al., 2020). For the PCL-5, two of the five models adequately fit the data which was based on: a borderline fit in some of the goodness-of-fit indicators (e.g., CMIN, CFI, RMSEA), and significant chi-square values (Hayduk, 2014a, 2014b). For the ITQ, the three-factor PTSD model and both the six-factor first-order and two-factor

second-order CPTSD models based on ICD-11 criteria showed the best fit to the data. Notably, Model 5 provided a statistically significant improvement in fit over Model 3. However, concerns about overfitting and multicollinearity arose in Model 5, with correlations exceeding 1.00 in the training set and original data set. These findings suggest that while Model 5 demonstrates superior statistical fit, Model 3 offers a more conceptually valid structure, as it avoids the issues of overfitting. It is important to caution that overfitting in Model 5 may indicate that the model is too specific to this data set and may not generalize well to other samples. Therefore, further testing across different data sets is required to ensure the robustness and applicability of this model. Overall, a good representation of the latent structure of the ITQ was found in this parent sample, with a trade-off between statistical fit and conceptual clarity.

This suggests that the ICD-11 models, even the worst among them, outperform the best DSM-5 models, indicating a superior representation of PTSD and CPTSD in this parent population. Further support for this finding comes from a subset analysis of participants meeting Criterion A for PTSD (n = 204), which also showed ICD-11 models demonstrating superior fit compared with DSM-5 models. Both of the best-fitting ITQ CPTSD models captured the distinction between PTSD and DSO symptoms inherent to the ICD-11 criteria among parents of autistic children and as a result were included in the analysis, despite an item loading of greater than 1 occurring in the affective dysregulation factor in the two-factor second-order model. As the error variance in this model was only slightly below zero, this item loading >1 may have been the result of sampling fluctuations and/or highly correlated factors, and not necessarily indicative of model misspecification (Cooperman & Waller, 2022; Kyriazos, 2018).

Further, the ITQ scale scores demonstrated acceptable levels of internal reliability, with the exception of affective dysregulation. This inadequate reliability was also observed by both Frost et al. (2022) and Sele et al. (2020), who attributed it to the presence of two opposing types of emotion regulation coping strategies (i.e., hyper- and hypo-affective states) one of which may be preferred more than the other. This theory provides a plausible explanation for the lower reliability observed in the affective dysregulation subscale in this study and aligns with existing literature on emotional regulation variability. However, further research is needed to clarify the role of these opposing coping strategies and their influence on the scale's reliability.

Despite this, all factor loadings for both models were positively and statistically significant, with mostly moderate to strong loadings, indicating that the relationships between the latent factors and the observed variables were substantial and statistically reliable. This implies that both models provide a close approximation of the sample data and represent viable models of the latent PTSD and CPTSD structure of the ITQ. Furthermore, indicating that the ITQ may be an acceptable standardized measurement tool to assess parenting experiences for PTSD and CPTSD specifically in parents of autistic children in both research and clinical contexts. This further underscores the robustness of the *ICD-11* criteria and the ITQ as superior tools for assessing PTSD and CPTSD in this parent population.

It was interesting to note that only two of the *DSM-5* models provided an adequate fit to the data given the broader symptom coverage that is inherent in this system (Hansen et al., 2017). The intrusion/distress model did show a superior fit of the more well-established *DSM-5* PTSD models; however, an overall good fit was still not demonstrated for this population. The fit of this model may have been impacted by lower standardized factor loadings occurring for two scale items (recklessness and trauma-related amnesia) and may suggest that item modification (e.g., parameter changes, and item adjustment) may improve the PCL-5 factor structure.

Our findings clearly indicate that the *ICD-11* PTSD criteria, as assessed by the ITQ, offer a more appropriate and reliable framework for trauma assessment and intervention in Australian parents of autistic children compared with the *DSM-5*. While these results align with the majority of findings from studies utilizing the ITQ/*ICD* framework, they also highlight unique aspects of trauma experienced by this specific population. The better fit of *ICD-11* models over *DSM-5* models may also be influenced by specific sociocultural factors related to our Australian sample. The ongoing caregiving stressors faced by parents of autistic children, such as social isolation, limited support services, and financial strain, might be particularly pronounced in Australia, potentially shaping how PTSD and CPTSD symptoms cluster in this population. The chronic and cumulative nature of these stressors aligns more closely with *ICD-11*'s broader framework, which distinguishes between PTSD and CPTSD. As such, this sample's unique experience may contribute to the observed superiority of *ICD-11* models, suggesting that sociocultural context plays an important role in shaping trauma responses in this parent population.

In particular, our results indicate that the self-report symptoms of PTSD do not appear to be represented by the traditional four-factor model (intrusion, avoidance, negative alterations in cognitions and mood, arousal and reactivity) in parents, and instead have a complexity that is best understood when looking at PTSD alongside

symptoms of DSO (e.g., feeling like a failure, worthless, distant or cutoff, and difficulties with emotional closeness). Trauma informed intervention has been reported to appropriately treat symptoms of DSO (e.g., Purnell et al., 2021). This may suggest the need for clinicians working with some parents of autistic children to utilize trauma-focused interventions. From a research perspective, future studies investigating the (Norris, 1990) mental health of this parent population may benefit from considering PTSD and CPTSD conceptualizations in addition to anxiety, depression, and stress.

Limitations and Future Research

A limitation of this study was the fact that only two posttraumatic stress measures were compared. Other scales (e.g., Traumatic Stress Schedule, Norris, 1990; and Traumatic Events Questionnaire, (Hayduk, 2014a, 2014b; Norris, 1990; Vrana & Lauterbach, 1994) might also provide adequate conceptualizations of PTSD and CPTSD in this parent population. Additionally, the use of a convenience sample may have introduced bias, limiting the generalizability of the findings.

The exclusion of the sexual assault item from the LEC-5. Although sexual assaults are highly correlated with PTSD, this item was omitted due to the minimal research connecting it to the experience of parenting trauma in raising autistic children. However, this omission may have influenced the overall results, as individuals who experienced sexual assault as their traumatic event might not have endorsed a Criterion A event.

The potential conflation of trauma, stress, and challenging events in the full sample was also a consideration. Further analyses isolating participants who met full diagnostic criteria for PTSD or CPTSD, including Criterion A for trauma exposure, demonstrated consistent psychometric properties within this subgroup. This finding addresses concerns regarding potential conflation and reinforces the robustness of our findings.

Furthermore, this study relied solely on self-report measures which can lead to discrepancies between reported symptoms those identified in clinical interviews. This reliance on self-report may have affected the reported prevalence and presentation of PTSD. Future research should consider complementing self-report measures with structured clinical interviews to provide a more comprehensive understanding of PTSD symptoms in this population.

Finally, it is important to highlight that this study did not assess the diagnostic accuracy of the PTSD measures used in this specific population. While tools provided valuable insights into PTSD and CPTSD prevalence, their accuracy in diagnosing these conditions among parents of autistic children remains untested.

Future studies should explore a broader range of trauma measures beyond the PCL-5 and ITQ to capture a more comprehensive understanding of PTSD and CPTSD in parents of autistic children. This could include scales such as the Traumatic Stress Schedule and the Traumatic Events Questionnaire. Additionally, to enhance the generalizability of findings, future research should employ random sampling methods rather than convenience sampling, ensuring a more representative sample of the parent population. Furthermore, it is crucial to further investigate the role of DSO symptoms in the manifestation of PTSD and CPTSD among parents of autistic children. The strong representation of DSO in the *ICD-11* models and the inadequacy of traditional PTSD models in capturing the complexity of symptoms in this population suggest that a deeper

understanding of DSO could inform the development of more targeted and effective therapeutic strategies.

Conclusion

The present study is the first to our knowledge to examine the fit of data from a population of Australian parents of autistic children to various *DSM-5* and *ICD-11* models of PTSD and CPTSD, respectively, as measured by the ITQ and the PCL-5. Empirical support was found for the three-factor *ICD-11* PTSD model, as well as the six-factor first-order and the two-factor second-order *ICD-11* CPTSD models when using the ITQ, suggesting that the factor structure of the ITQ may be a valid screening assessment for PTSD and CPTSD in this parent population. Additional analysis of Models 3 and 5 revealed that Model 5 provided a significantly better statistical fit; however, concerns over multicollinearity and overfitting, indicated by correlations exceeding 1.00 in the training set and original data set, suggesting that Model 3 may offer a more conceptually valid structure. While Model 3 captures a better statistical fit, the trade-off with Model 3's conceptual clarity should be considered. Two of the *DSM-5* models tested using the PCL-5 showed an adequate fit to the data with the remaining three models providing a poor fit. The novel intrusion/distress model showed a superior fit compared with the other four *DSM-5* models tested. While the results support the factorial validity and internal reliability of both the PCL-5 and the ITQ, the ITQ may provide a more valid PTSD and CPTSD measure for this parent population.

References

- Armour, C., Múllerová, J., & Elhai, J. D. (2016). A systematic literature review of PTSD's latent structure in the Diagnostic and Statistical Manual of Mental Disorders: *DSM-IV* to *DSM-5*. *Clinical Psychology Review*, 44, 60–74. <https://doi.org/10.1016/j.cpr.2015.12.003>
- Armour, C., Robinson, M., & Ross, J. (2021). Factor structure of the International Trauma Questionnaire in UK Armed Forces veterans residing in Northern Ireland. *European Journal of Psychotraumatology*, 12(1), Article 1924954. <https://doi.org/10.1080/20008198.2021.1924954>
- Armour, C., Tsai, J., Durham, T. A., Charak, R., Biehn, T. L., Elhai, J. D., & Pietrzak, R. H. (2015). Dimensional structure of *DSM-5* posttraumatic stress symptoms: Support for a hybrid Anhedonia and Externalizing Behaviors model. *Journal of Psychiatric Research*, 61, 106–113. <https://doi.org/10.1016/j.jpsychires.2014.10.012>
- Blevins, C. A., Weathers, F. W., Davis, M. T., Witte, T. K., & Domino, J. L. (2015). The Posttraumatic Stress Disorder Checklist for *DSM-5* (PCL-5): Development and Initial Psychometric Evaluation. *Journal of Traumatic Stress*, 28(6), 489–498. <https://doi.org/10.1002/jts.22059>
- Bollen, K. A., & Stine, R. A. (1992). Bootstrapping goodness-of-fit measures in structural equation models. *Sociological Methods & Research*, 21(2), 205–229. <https://doi.org/10.1177/0049124192021002004>
- Botha, M., Chapman, R., Giwa Onaiwu, M., Kapp, S. K., Stannard Ashley, A., & Walker, N. (2024). The neurodiversity concept was developed collectively: An overdue correction on the origins of neurodiversity theory. *Autism*, 28(6), 1591–1594. <https://doi.org/10.1177/13623613241237871>
- Bovin, M. J., Marx, B. P., Weathers, F. W., Gallagher, M. W., Rodriguez, P., Schnurr, P. P., & Keane, T. M. (2016). Psychometric properties of the PTSD Checklist for Diagnostic and Statistical Manual of Mental Disorders-Fifth Edition (PCL-5) in veterans. *Psychological Assessment*, 28(11), 1379–1391. <https://doi.org/10.1037/pas0000254>
- Bury, S. M., Jellett, R., Spoor, J. R., & Hedley, D. (2023). "It defines who i am" or "It's something i have": What language do [Autistic] Australian adults [on the Autism Spectrum] prefer? *Journal of Autism and Developmental Disorders*, 53(2), 677–687. <https://doi.org/10.1007/s10803-020-04425-3>
- Cloitre, M., Garvert, D. W., Weiss, B., Carlson, E. B., & Bryant, R. A. (2014). Distinguishing PTSD, Complex PTSD, and Borderline Personality Disorder: A latent class analysis. *European Journal of Psychotraumatology*, 5(1), Article 25097. <https://doi.org/10.3402/ejpt.v5.25097>
- Cloitre, M., Hyland, P., Bisson, J. I., Brewin, C. R., Roberts, N. P., Karatzias, T., & Shevlin, M. (2019). *ICD-11* Posttraumatic Stress Disorder and Complex Posttraumatic Stress Disorder in the United States: A Population-Based Study. *Journal of Traumatic Stress*, 32(6), 833–842. <https://doi.org/10.1002/jts.22454>
- Cooperman, A. W., & Waller, N. G. (2022). Heywood you go away! Examining causes, effects, and treatments for Heywood cases in exploratory factor analysis. *Psychological Methods*, 27(2), 156–176. <https://doi.org/10.1037/met0000384>
- Dhingra, K., Mitchell, S. M., Davies, B., & Taylor, P. J. (2021). Examining the factor structure of *ICD-11* posttraumatic stress disorder (PTSD) and complex-PTSD among prison staff exposed to potentially traumatic experiences. *Psychiatry Research*, 303, Article 114085. <https://doi.org/10.1016/j.psychres.2021.114085>
- Elklit, A., Hyland, P., & Shevlin, M. (2014). Evidence of symptom profiles consistent with posttraumatic stress disorder and complex posttraumatic stress disorder in different trauma samples. *European Journal of Psychotraumatology*, 5(1), Article 24221. <https://doi.org/10.3402/ejpt.v5.24221>
- Elliott, R., McKinnon, A., Dixon, C., Boyle, A., Murphy, F., Dahm, T., Travers-Hill, E., Mul, C. L., Archibald, S. J., Smith, P., Dalgleish, T., Meiser-Stedman, R., & Hitchcock, C. (2021). Prevalence and predictive value of *ICD-11* post-traumatic stress disorder and Complex PTSD diagnoses in children and adolescents exposed to a single-event trauma. *Journal of Child Psychology and Psychiatry*, 62(3), 270–276. <https://doi.org/10.1111/jcpp.13240>
- Feigin, H., Baror, S., Bar, M., & Zaidel, A. (2021). Perceptual decisions are biased toward relevant prior choices. *Scientific Reports*, 11(1), Article 648. <https://doi.org/10.1038/s41598-020-80128-0>
- Forbes, D., Lockwood, E., Elhai, J. D., Creamer, M., Bryant, R., McFarlane, A., Silove, D., Miller, M. W., Nickerson, A., & O'Donnell, M. (2015). An evaluation of the *DSM-5* factor structure for posttraumatic stress disorder in survivors of traumatic injury. *Journal of Anxiety Disorders*, 29, 43–51. <https://doi.org/10.1016/j.janxdis.2014.11.004>
- Frost, R., Vang, M. L., Hyland, P., Shevlin, M., McCarthy, A., & Murphy, J. (2022). Assessing the factorial validity and the internal reliability of the International Trauma Questionnaire (ITQ); PTSD and complex PTSD among survivors of sexual violence in Ireland. *Epidemiology and Psychiatric Sciences*, 31, Article e42. <https://doi.org/10.1017/S2045796022000245>
- Gilbar, O., Hyland, P., Cloitre, M., & Dekel, R. (2018). *ICD-11* complex PTSD among Israeli male perpetrators of intimate partner violence: Construct validity and risk factors. *Journal of Anxiety Disorders*, 54, 49–56. <https://doi.org/10.1016/j.janxdis.2018.01.004>
- Gilmour, J., & Romaniuk, M. (2020). Factor structure of posttraumatic stress disorder (PTSD) in Australian Vietnam Veterans: Confirmatory factor analysis of the Clinician-Administered PTSD Scale for *DSM-5*. *Journal of Military, Veteran and Family Health*, 6(1), 48–57. <https://doi.org/10.3138/jmfvh-2018-0042>
- Glück, T. M., Knefel, M., Tran, U. S., & Lueger-Schuster, B. (2016). PTSD in *ICD-10* and proposed *ICD-11* in elderly with childhood trauma: prevalence, factor structure, and symptom profiles. *European Journal of Psychotraumatology*, 7(1), Article 29700. <https://doi.org/10.3402/ejpt.v7.29700>
- Hansen, M., Hyland, P., Armour, C., Shevlin, M., & Elklit, A. (2015). Less is more? Assessing the validity of the *ICD-11* model of PTSD across multiple trauma samples. *European Journal of Psychotraumatology*, 6(1), Article 28766. <https://doi.org/10.3402/ejpt.v6.28766>

- Hansen, M., Hyland, P., Karstoft, K. I., Vaegter, H. B., Bramsen, R. H., Nielsen, A. B. S., Armour, C., Andersen, S. B., Høybye, M. T., Larsen, S. K., & Andersen, T. E. (2017). Does size really matter? A multisite study assessing the latent structure of the proposed *ICD-11* and *DSM-5* diagnostic criteria for PTSD. *European Journal of Psychotraumatology*, 8(sup7), Article 1398002. <https://doi.org/10.1080/20008198.2017.1398002>
- Hansen, M., Vaegter, H. B., Cloitre, M., & Andersen, T. E. (2021). Validation of the Danish International Trauma Questionnaire for posttraumatic stress disorder in chronic pain patients using clinician-rated diagnostic interviews. *European Journal of Psychotraumatology*, 12(1), Article 1880747. <https://doi.org/10.1080/20008198.2021.1880747>
- Haravuori, H., Kiviruusu, O., Suomalainen, L., & Marttunen, M. (2016). An evaluation of *ICD-11* posttraumatic stress disorder criteria in two samples of adolescents and young adults exposed to mass shootings: Factor analysis and comparisons to *ICD-10* and *DSM-IV*. *BMC Psychiatry*, 16(1), Article 140. <https://doi.org/10.1186/s12888-016-0849-y>
- Hayduk, L. A. (2014a). Seeing perfectly fitting factor models that are causally misspecified: Understanding that close-fitting models can be worse. *Educational and Psychological Measurement*, 74(6), 905–926. <https://doi.org/10.1177/0013164414527449>
- Hayduk, L. A. (2014b). Shame for disrespecting evidence: The personal consequences of insufficient respect for structural equation model testing. *BMC Medical Research Methodology*, 14, Article 124. <https://doi.org/10.1186/1471-2288-14-124>
- Hu, L., & Bentler, P. M. (1999). cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, 6(1), 1–55. <https://doi.org/10.1080/10705519909540118>
- Hyland, P., Murphy, J., Shevlin, M., Vallières, F., McElroy, E., Elklit, A., Christoffersen, M., & Cloitre, M. (2017). Variation in post-traumatic response: The role of trauma type in predicting *ICD-11* PTSD and CPTSD symptoms. *Social Psychiatry and Psychiatric Epidemiology*, 52(6), 727–736. <https://doi.org/10.1007/s00127-017-1350-8>
- Islam, M. S., Ferdous, M. Z., Sujon, M. S. H., Tasnim, R., Masud, J. H. B., Kundu, S., Mosaddek, A. S. M., Choudhuri, M. S. K., Kira, I. A., & Gozal, D. (2021). The psychometric properties of the Bangla Posttraumatic Stress Disorder Checklist for DSM-5 (PCL-5): Preliminary reports from a large-scale validation study. *BMC Psychiatry*, 22, Article 280. <https://doi.org/10.1186/s12888-022-03920-4>
- Izuno-Garcia, A. K., McNeel, M. M., & Fein, R. H. (2023). Neurodiversity in promoting the well-being of children on the autism spectrum. *Child Care in Practice*, 29(1), 54–67. <https://doi.org/10.1080/13575279.2022.2126436>
- Karatzias, T., Shevlin, M., Ben-Ezra, M., McElroy, E., Redican, E., Vang, M. L., Cloitre, M., Ho, G. W. K., Lorberg, B., Martsenkovskiy, D., & Hyland, P. (2023). War exposure, posttraumatic stress disorder, and complex posttraumatic stress disorder among parents living in Ukraine during the Russian war. *Acta Psychiatrica Scandinavica*, 147(3), 276–285. <https://doi.org/10.1111/acps.13529>
- Karatzias, T., Shevlin, M., Fyvie, C., Hyland, P., Efthymiadou, E., Wilson, D., Roberts, N., Bisson, J. I., Brewin, C. R., & Cloitre, M. (2016). An initial psychometric assessment of an *ICD-11* based measure of PTSD and complex PTSD (*ICD-TQ*): Evidence of construct validity. *Journal of Anxiety Disorders*, 44, 73–79. <https://doi.org/10.1016/j.janxdis.2016.10.009>
- Kline, R. B. (1999). Book review: Psychometric theory. *Journal of Psychoeducational Assessment*, 17(3), 275–280. <https://doi.org/10.1177/073428299901700307>
- Kline, R. B. (2022). *Principles and practice of structural equation modelling* (5th ed.). Guilford Press.
- Krüger-Gottschalk, A., Ehling, T., Knaevelsrud, C., Dyer, A., Schäfer, I., Schellong, J., Rau, H., & Köhler, K. (2022). Confirmatory factor analysis of the Clinician-Administered PTSD Scale (CAPS-5) based on *DSM-5* vs. *ICD-11* criteria. *European Journal of Psychotraumatology*, 13(1), Article 2010995. <https://doi.org/10.1080/20008198.2021.2010995>
- Kyriazos, T. (2018). Applied psychometrics: Sample size and sample power considerations in factor analysis (EFA, CFA) and SEM in general. *Psychology*, 9(8), 2207–2230. <https://doi.org/10.4236/psych.2018.98126>
- La Greca, A. M., Danzi, B. A., & Chan, S. F. (2017). *DSM-5* and *ICD-11* as competing models of PTSD in preadolescent children exposed to a natural disaster: Assessing validity and co-occurring symptomatology. *European Journal of Psychotraumatology*, 8(1), Article 1310591. <https://doi.org/10.1080/20008198.2017.1310591>
- Lee, D. J., Bovin, M. J., Weathers, F. W., Palmieri, P. A., Schnurr, P. P., Sloan, D. M., Keane, T. M., & Marx, B. P. (2019). Latent factor structure of *DSM-5* posttraumatic stress disorder: Evaluation of method variance and construct validity of novel symptom clusters. *Psychological Assessment*, 31(1), 46–58. <https://doi.org/10.1037/pas0000642>
- Maercker, A., Brewin, C. R., Bryant, R. A., Cloitre, M., Reed, G. M., van Ommeren, M., Humayun, A., Jones, L. M., Kagee, A., Llosa, A. E., Rousseau, C., Somasundaram, D. J., Souza, R., Suzuki, Y., Weissbecker, I., Wessely, S. C., First, M. B., & Saxena, S. (2013). Proposals for mental disorders specifically associated with stress in the International Classification of Diseases-11. *The Lancet*, 381(9878), 1683–1685. [https://doi.org/10.1016/S0140-6736\(12\)62191-6](https://doi.org/10.1016/S0140-6736(12)62191-6)
- Martin, C. J. H., Patterson, J., Paterson, C., Welsh, N., Dougall, N., Karatzias, T., & Williams, B. (2021). *ICD-11* complex post traumatic stress disorder (CPTSD) in parents with perinatal bereavement: Implications for treatment and care. *Midwifery*, 96, Article 102947. <https://doi.org/10.1016/j.midw.2021.102947>
- Mordeno, I. G., Nalipay, M. J. N., & Mordeno, E. R. (2019). The factor structure of complex PTSD in combat-exposed Filipino soldiers. *Psychiatry Research*, 278, 65–69. <https://doi.org/10.1016/j.psychres.2019.05.035>
- Murphy, D., Shevlin, M., Pearson, E., Greenberg, N., Wessely, S., Busuttil, W., & Karatzias, T. (2020). A validation study of the International Trauma Questionnaire to assess post-traumatic stress disorder in treatment-seeking veterans. *The British Journal of Psychiatry*, 216(3), 132–137. <https://doi.org/10.1192/bjp.2020.9>
- Murphy, S., Elklit, A., Dokkedahl, S., & Shevlin, M. (2018). Testing competing factor models of the latent structure of post-traumatic stress disorder and complex post-traumatic stress disorder according to *ICD-11*. *European Journal of Psychotraumatology*, 9(1), Article 1457393. <https://doi.org/10.1080/20008198.2018.1457393>
- Norris, F. H. (1990). Screening for traumatic stress: A scale for use in the general population. *Journal of Applied Social Psychology*, 20(20), 1704–1718. <https://doi.org/10.1111/j.1559-1816.1990.tb01505.x>
- Purnell, L. R., Graham, A. C. J., Bloomfield, M. A. P., & Billings, J. (2021). Reintegration interventions for CPTSD: A systematic review. *European Journal of Psychotraumatology*, 12(1), Article 1934789. <https://doi.org/10.1080/20008198.2021.1934789>
- Rahimi, M. P., Wafa, M. H., Stanikzai, M. H., & Rahimi, B. A. (2023). Post-traumatic stress disorder (PTSD) probability among parents who live in Kandahar, Afghanistan and lost at least a child to armed conflict. *Scientific Reports*, 13(1), Article 3994. <https://doi.org/10.1038/s41598-023-31228-0>
- Redican, E., Nolan, E., Hyland, P., Cloitre, M., McBride, O., Karatzias, T., Murphy, J., & Shevlin, M. (2021). A systematic literature review of factor analytic and mixture models of *ICD-11* PTSD and CPTSD using the International Trauma Questionnaire. *Journal of Anxiety Disorders*, 79, Article 102381. <https://doi.org/10.1016/j.janxdis.2021.102381>
- Sele, P., Hoffart, A., Bækkelund, H., & Øktdalen, T. (2020). Psychometric properties of the International Trauma Questionnaire (ITQ) examined in a Norwegian trauma-exposed clinical sample. *European Journal of Psychotraumatology*, 11(1), Article 1796187. <https://doi.org/10.1080/20008198.2020.1796187>
- Seo, W., & Cho, Y. (2023). The psychometric properties of the posttraumatic stress disorder checklist for the *DSM-5* in Korean adults exposed to natural disasters. *Japanese Psychological Research*, 65(1), 75–84. <https://doi.org/10.1111/jpr.12341>

- Shevlin, M., Hyland, P., Karatzias, T., Fyvie, C., Roberts, N., Bisson, J. I., Brewin, C. R., & Cloitre, M. (2017). Alternative models of disorders of traumatic stress based on the new *ICD-11* proposals. *Acta Psychiatrica Scandinavica*, 135(5), 419–428. <https://doi.org/10.1111/acps.12695>
- Silove, D., Tay, A. K., Kareth, M., & Rees, S. (2017). The relationship of complex post-traumatic stress disorder and post-traumatic stress disorder in a culturally distinct, conflict-affected population: A study among west Papuan refugees displaced to Papua New Guinea. *Frontiers in Psychiatry*, 8, Article 73. <https://doi.org/10.3389/fpsy.2017.00073>
- Soberón, C., Crespo, M., del Mar Gómez-Gutiérrez, M., Fernández-Lansac, V., & Armour, C. (2016). Dimensional structure of DSM-5 posttraumatic stress symptoms in Spanish trauma victims. *European Journal of Psychotraumatology*, 7(1), Article 32078. <https://doi.org/10.3402/ejpt.v7.32078>
- Somma, A., Maffei, C., Borroni, S., Gialdi, G., & Fossati, A. (2019). Post traumatic reactions as individual differences: Latent structure analysis of the International Trauma Questionnaire in Italian trauma-exposed and non-trauma exposed adults. *Mediterranean Journal of Clinical Psychology*, 7(1). <https://doi.org/10.6092/2282-1619/2019.7.2058>
- Stephenson, K. G., Fenning, R. M., Macklin, E. A., Lu, F., Norris, M., Steinberg-Epstein, R., & Butter, E. M. (2023). Child behavior problems and parenting stress in underserved families of children with ASD: Investigation of family resources and parenting self-efficacy. *Journal of Autism and Developmental Disorders*, 53(10), 3787–3798. <https://doi.org/10.1007/s10803-022-05681-1>
- Stewart, M., Schnabel, A., Hallford, D. J., McGillivray, J. A., Forbes, D., Foster, M., Shandley, K., Gardam, M., & Austin, D. W. (2020). Challenging child behaviours positively predict symptoms of posttraumatic stress disorder in parents of children with autism spectrum disorder and rare diseases. *Research in Autism Spectrum Disorders*, 69, Article 101467. <https://doi.org/10.1016/j.rasd.2019.101467>
- Sveen, J., Bondjers, K., & Willebrand, M. (2016). Psychometric properties of the PTSD Checklist for *DSM-5*: A pilot study. *European Journal of Psychotraumatology*, 7(1), Article 30165. <https://doi.org/10.3402/ejpt.v7.30165>
- Taboas, A., Doepke, K., & Zimmerman, C. (2023). Preferences for identity-first versus person-first language in a US sample of autism stakeholders. *Autism*, 27(2), 565–570. <https://doi.org/10.1177/13623613221130845>
- Tay, A. K., Mohsin, M., Rees, S., Tam, N., Kareth, M., & Silove, D. (2018). Factor structures of complex posttraumatic stress disorder and PTSD in a community sample of refugees from West Papua. *Comprehensive Psychiatry*, 85, 15–22. <https://doi.org/10.1016/j.comppsy.2018.05.001>
- Tay, A. K., Rees, S., Chen, J., Kareth, M., & Silove, D. (2015). The structure of post-traumatic stress disorder and complex post-traumatic stress disorder amongst West Papuan refugees. *BMC Psychiatry*, 15(1), Article 111. <https://doi.org/10.1186/s12888-015-0480-3>
- Vallièrès, F., Ceannt, R., Daccache, F., Abou Daher, R., Sleiman, J., Gilmore, B., Byrne, S., Shevlin, M., Murphy, J., & Hyland, P. (2018). *ICD-11* PTSD and complex PTSD amongst Syrian refugees in Lebanon: The factor structure and the clinical utility of the International Trauma Questionnaire. *Acta Psychiatrica Scandinavica*, 138(6), 547–557. <https://doi.org/10.1111/acps.12973>
- Vrana, S., & Lauterbach, D. (1994). Prevalence of traumatic events and post-traumatic psychological symptoms in a nonclinical sample of college students. *Journal of Traumatic Stress*, 7(2), 289–302. <https://doi.org/10.1007/BF02102949>
- Wang, Q., Xu, W., Ren, L., Wang, W., & Wang, Y. (2019). The relationship between hope and post-traumatic stress disorder in Chinese Shidu parents: The mediating role of perceived stress. *Journal of Affective Disorders*, 251, 23–30. <https://doi.org/10.1016/j.jad.2019.03.049>
- Weathers, F. W., Blake, D. D., Schnurr, P. P., Kaloupek, D. G., Marx, B. P., & Keane, T. M. (2013). *The Life Events Checklist for DSM-5 (LEC-5)*. Instrument available from the National Center for PTSD. <https://www.ptsd.va.gov>
- West, S. G., Taylor, A. B., & Wu, W. (2012). Model fit and model selection in structural equation modeling. In R. H. Hoyle (Ed.), *Handbook of structural equation modeling* (pp. 209–231). The Guilford Press.
- World Health Organisation. (2018). *International statistical classification of diseases and related health problems 11th revision*. <https://www.who.int/classifications/icd/revision/en/>
- Wortmann, J. H., Jordan, A. H., Weathers, F. W., Resick, P. A., Dondanville, K. A., Hall-Clark, B., Foa, E. B., Young-McCaughan, S., Yarvis, J. S., Hembree, E. A., Mintz, J., Peterson, A. L., & Litz, B. T. (2016). Psychometric analysis of the PTSD Checklist-5 (PCL-5) among treatment-seeking military service members. *Psychological Assessment*, 28(11), 1392–1403. <https://doi.org/10.1037/pas0000260>
- Xiong, T., McGrath, P. J., Stewart, S. H., Bagnell, A., & Kaltenbach, E. (2022). Risk and protective factors for posttraumatic stress and posttraumatic growth in parents of children with intellectual and developmental disorders. *European Journal of Psychotraumatology*, 13(1), Article 2087979. <https://doi.org/10.1080/20008198.2022.2087979>
- Xiong, T., McGrath, P. J., Yakovenko, I., Thomson, D., & Kaltenbach, E. (2022). Parenting-related trauma exposure among parents of children with intellectual and developmental disorders: Development and validation of the parenting trauma checklist. *Journal of Traumatic Stress*, 35(2), 759–770. <https://doi.org/10.1002/jts.22779>
- Yang, H., Wang, L., Cao, C., Cao, X., Fang, R., Zhang, J., & Elhai, J. D. (2017). The underlying dimensions of DSM-5 PTSD symptoms and their relations with anxiety and depression in a sample of adolescents exposed to an explosion accident. *European Journal of Psychotraumatology*, 8(1), Article 1272789. <https://doi.org/10.1080/20008198.2016.1272789>

Received July 13, 2024

Revision received March 13, 2025

Accepted April 17, 2025 ■